STAT/MA 41600

Midterm Exam #1: October 9, 2026

Name _____

Purdue student ID (10 digits) _____

1. The testing booklet contains 5 questions. Dr. Ward will grade them all, but will only use the highest 4 scores. These highest 4 scores will all be weighted evenly (i.e., each question is worth 1/4 of the midterm exam grade).

2. Permitted Texas Instruments calculators:
   BA-35
   BA II Plus*
   BA II Plus Professional Edition*
   TI-30XS MultiView*
   TI-30Xa
   TI-30XIIS*
   TI-30XIIB*
   TI-30XB MultiView*
   *The memory of the calculator should be cleared at the start of the exam.

3. **Circle your final answer in your booklet**; otherwise, no credit may be given.

4. There is no penalty for guessing or partial work.

5. Show all your work in the exam booklet. If the majority of questions are answered correctly, but insufficient work is given, the exam could be considered for academic misconduct.

"As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together—We are Purdue."

FIRST VERSION

1a. Suppose that, during a given week, 5,000,000 people play a lottery game. If their chances to win the lottery are independent, and if each person has probability 1/2,000,000 of winning the lottery, write an *exact expression* for the probability that there are exactly 4 winners of the lottery that week. (You do not need to find the decimal answer in part **a**.)

1b. Briefly explain how you can approximate the value in part **a** using a Poisson random variable. Then give an approximate value for the probability that there are exactly 4 winners.

SECOND VERSION

1a. Dr. Ward estimates that, among 50,000 students on campus, the students choose independently whether to visit the MATH library on a given day. Each student decides to visit with probability 1/10,000, or to avoid the library with probability 9,999/10,000. Let X denote the number of people who visit the MATH library on a given day. Write an *exact expression* for the probability that there are exactly 4 visitors to the MATH library on a given day. (You do not need to find the decimal answer in part **a**.)

1b. Briefly explain how you can approximate the value in part **a** using a Poisson random variable. Then give an approximate value for the probability that there are exactly 4 visitors.

FIRST VERSION

2. Rhonda rolls a fair 6-sided die until the first occurrence of "2", and then she stops afterwards. Bernadette rolls a fair 4-sided die until the first occurrence of "2", and then she stops afterwards. (Assume that all rolls are independent.)

2a. What is the probability that the number of Rhonda's rolls is the same as the number of Bernadette's rolls?

2b. What is the probability that the number of Rhonda's rolls strictly exceeds the number of Bernadette's rolls?

SECOND VERSION

2. Rhonda rolls a fair 8-sided die until the first occurrence of "2", and then she stops afterwards. Bernadette flips a coin until the first occurrence of "heads", and then she stops afterwards. (Assume that all rolls and flips are independent.)

2a. What is the probability that the number of Rhonda's rolls is the same as the number of Bernadette's flips?

2b. What is the probability that the number of Rhonda's rolls strictly exceeds the number of Bernadette's flips?

FIRST VERSION

3. In my sock drawer there are 21 white socks, 8 black socks, and 4 brown socks. (The socks are *not* folded into pairs.) I pick 6 socks at random, without replacement. All selections of 6 socks are equally likely.

The word "match" in this problem means "share the same color".

3a. What is the probability that I pull out exactly 2 matching socks of each color?

3b. What is the probability all 6 socks match?

3c. What is the probability that I pick exactly 4 white socks (the other 2 are not white)?

SECOND VERSION

3. In my sock drawer there are 18 white socks, 11 black socks, and 4 brown socks. (The socks are *not* folded into pairs.) I pick 6 socks at random, without replacement. All selections of 6 socks are equally likely.

The word "match" in this problem means "share the same color".

3a. What is the probability that I pull out exactly 2 matching socks of each color?

3b. What is the probability all 6 socks match?

3c. What is the probability that I pick exactly 4 white socks (the other 2 are not white)?

FIRST VERSION

4. Suppose that 42% of the pages of an analysis textbook have a theorem, 25% of the pages of an algebra textbook have a theorem, and 15% of the pages of a probability textbook have a theorem. The academic advisors have encouraged students to take at most one math course at a time, and as a result, 7% of students are taking analysis, 15% are taking algebra, 18% are taking probability, and the other 60% are not taking a math course. If you randomly walk up to a table at the local coffee shop and peek over the student's shoulder, and the student happens to be reading a theorem in a math book, what is the probability that student is enrolled in a probability course?

SECOND VERSION

4. The genres of the songs on a student's iPod are: 10% country music, 20% pop, 25% R&B, and 45% rock. Suppose that 92% of the country music songs have a fiddle, 16% of the pop songs have a fiddle, none of the R&B songs have a fiddle, and 12% of the rock songs have a fiddle. If a song is randomly selected from the iPod, and it happens to have fiddle music in the song, what is the conditional probability that it is a country song?

FIRST VERSION

5. Consider a collection of 30 bears, namely, 10 red bears, 10 green bears, and 10 blue bears. Select 9 of the bears (without replacement), and let X denote the number of bears that are red or blue.

5a. What is the expected value of X?

5b. What is the variance of X?

SECOND VERSION

5. Consider a collection of 30 bears, namely, 10 red bears, 10 green bears, and 10 blue bears. Select 8 of the bears (without replacement), and let X denote the number of bears that are red or blue.

5a. What is the expected value of X?

5b. What is the variance of X?